

ON NEUROMORPHIC COMPUTING

WITH SPIKING NEURAL NETWORKS

08.06.2026

INTRODUCTION AND MOTIVATION

- Deep learning has exceeded human performance in some domains and specific problems — AlphaFold, GPT, AlphaGo [1], [2]
- Yet training state-of-the-art models consumes vast energy and is rapidly approaching physical and economic limits [3], [4], [5], [6], [7]
- The human brain operates on 20 W and still outperforms AI on adaptation, few-shot learning, and common-sense reasoning [8]

Let's draw more inspiration from biological intelligence → **neuromorphic computing**

Key takeaway lessons from the brain and what we should base our neuromorphic architecture on:

- **Sparse:** Only sends and computes when needed — silence is free
- **Event driven:** Highly efficient neuron communication via binary spikes; information encoded in timing, not magnitude
- **Local:** Neurons hold their own weights; compute and memory are co-located, physically in the same place (eliminating the von Neumann bottleneck)

RESEARCH OBJECTIVES

Sparse Efficient Computing: Investigate whether biologically inspired, event-driven algorithms reduce the computational footprint of visual classification when simulated on standard hardware. **Neuron Model Evaluation:** Identify which temporal integration dynamics are compatible with Time-To-First-Spike (TTFS) rank-order decoding across systematic threshold regimes. **Inference Via Weight Transfer:** Quantify the accuracy penalty of zero-shot Artificial-Neural-Network (ANN) to Spiking-Neural-Network (SNN) weight transfer under TTFS encoding, isolating the cost of transitioning to event-driven integration. **Native Unsupervised Learning:** Determine if local Spike-Time-Dependent-Plasticity (STDP) can autonomously extract meaningful geometric features from visual input without global error signals.

First let's look at how the brain works...

NEURON STRUCTURE & FUNCTION

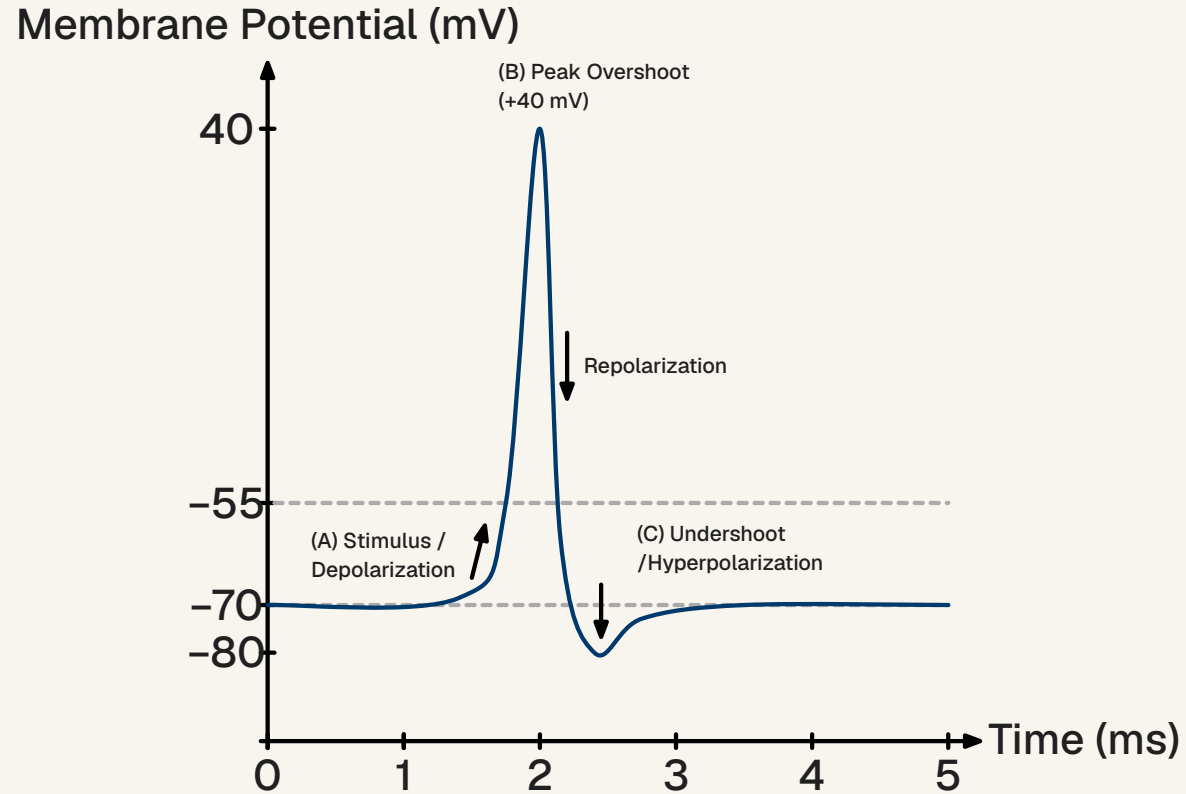
A biological neuron is a highly specialized cell that is the foundational building block in the nervous system. Functionally it is a time-dependent, event-driven processor with three functional zones:

- **Dendrites (Input):** Branching tree collecting signals from thousands of upstream neurons via synaptic terminals
- **Soma (Integration):** Cell body summing competing excitatory and inhibitory inputs; acts as a biological capacitor
- **Axon (Output):** Long cable transmitting the neuron's output spike to downstream targets; insulated by myelin for fast conduction

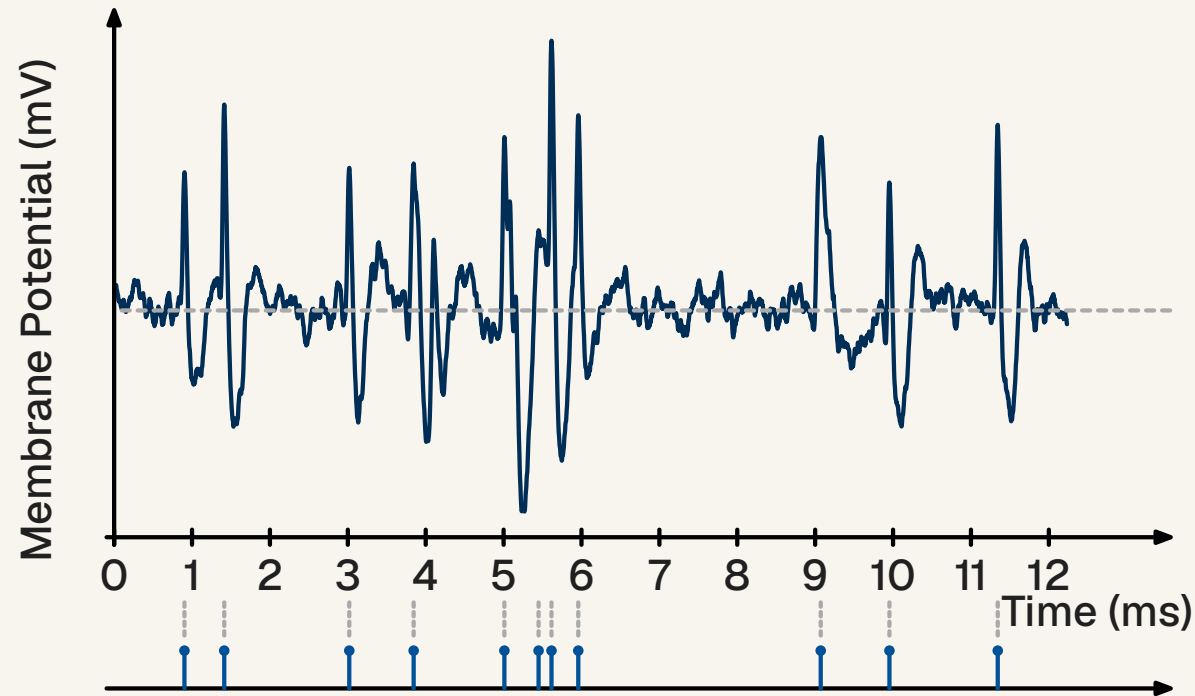
[9], [10]

All-or-nothing principle: If aggregate membrane potential exceeds the threshold, voltage-gated channels cascade open → rapid depolarisation spike. Sub-threshold noise is silently discarded. The neuron then enters a **refractory period**, resetting the gradient.

ACTION POTENTIAL & SPIKE TRAINS



Information is encoded entirely in the **precise timing** of spikes — abstracted as a sum of Dirac delta functions: $S(t) = \sum_f \delta(t - t_f)$



Because the spike waveform is stereotypical and invariant across neurons, **the waveform itself carries no information.** [10], [11]

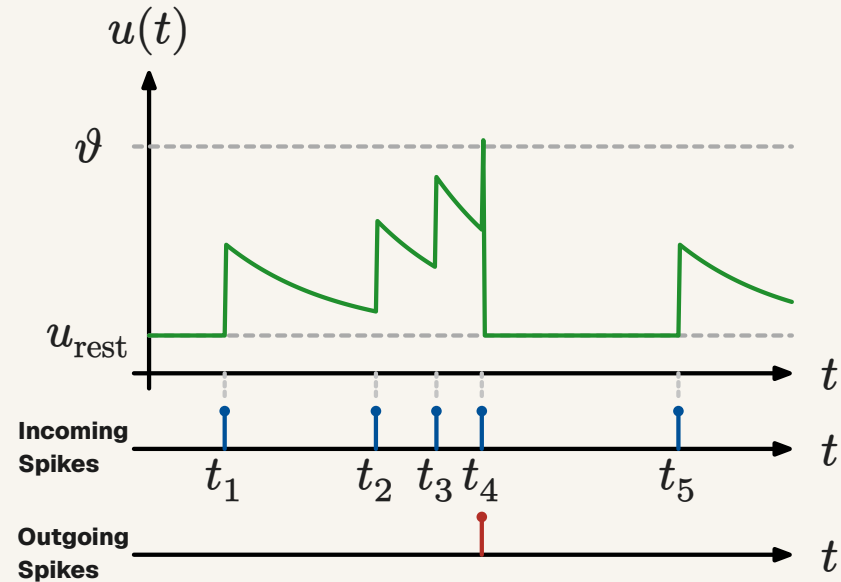
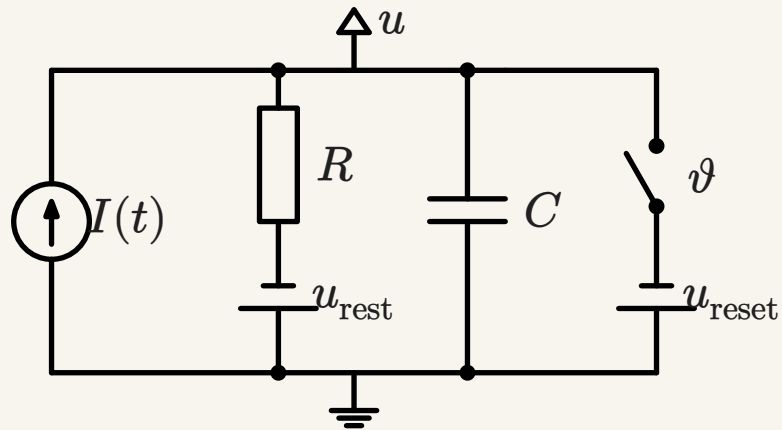
NEURON MODELS

A fundamental trade-off: **biological realism** vs **computational efficiency**

- **Hodgkin-Huxley:** Full ionic channel dynamics — accurate, but requires supercomputing for large networks
- **Integrate-and-Fire (IF):** Pure arithmetic accumulator; fires when $u > \vartheta$, then resets. Minimal overhead — $O(1)$ per spike.
- **Leaky Integrate-and-Fire (LIF):** Adds exponential membrane decay τ_m :
- **Generalised LIF (GLIF):** Adds an adaptation variable $w(t)$ enabling bursting, frequency adaptation, and richer firing patterns

[12], [10]

LIF



The LIF forms the foundation of the other neuron models in this thesis. The main properties of the LIF are:

integration, firing when reaching threshold, leaking

ENCODING INFORMATION IN SPIKES

The brain occupies a unique middle ground between digital and analog:

- Membrane potential is **analog** — continuous voltage
- Spike output is **binary** — discrete, stereotypical events in continuous time

This creates a **quantised continuous signal**: discrete amplitude, continuous time.

Requires a suitable codec to be useful as a communication medium.

Two primary coding schemes are hypothesised...

RATE ENCODING

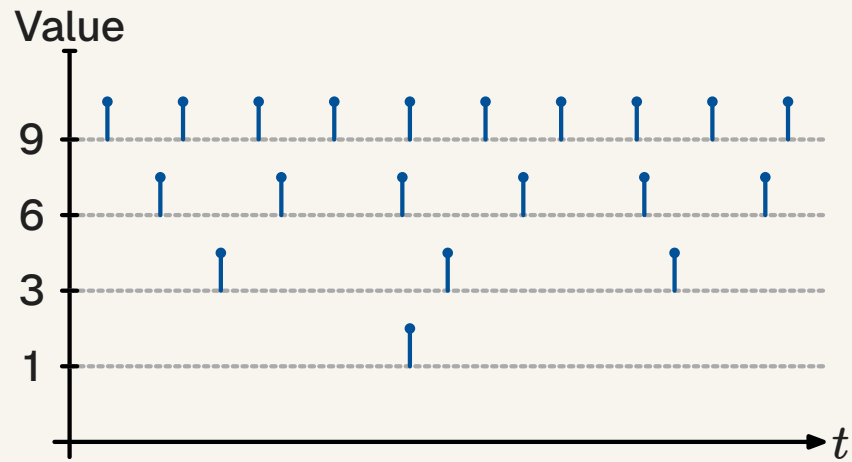
- Stimulus intensity is encoded as **mean firing frequency** over a time window — stronger stimulus → more spikes per second
- Simple and robust; directly observed in motor and sensory neurons
- **Limitation — latency barrier:** The post-synaptic neuron must integrate spikes over tens–hundreds of milliseconds to estimate a reliable rate
- This contradicts biological reaction times often under 100 ms, suggesting rate coding alone cannot account for time-critical processing [13]
- On digital hardware, high firing rates cause rapid transistor switching — costly for power draw and bus congestion

[14], [10]

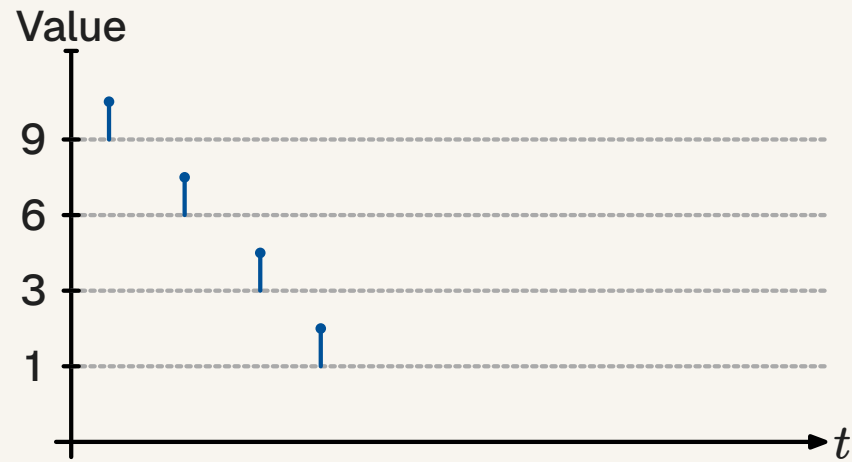
TIME-TO-FIRST-SPIKE (TTFS) ENCODING

- Unlike Rate Coding, which requires extended time windows, TTFS encodes stimulus intensity inversely to response latency
- **Mechanism:** A high-intensity (bright) pixel triggers an early spike, compressing spatial information into a priority-driven queue
- **Advantages:**
 - **Sparsity:** Sub-threshold noise (background pixels) is aggressively discarded and never fires
 - **Early Exit:** Salient features arrive first, enabling confident decisions within the first 15% of the temporal window
 - **Latency:** Eliminates the need to wait for a time window to close; processing terminates as soon as the threshold is reached
- Phase ambiguity is resolved by simulating a biological saccade, establishing a global temporal reference frame (t_0)

[15], [10]

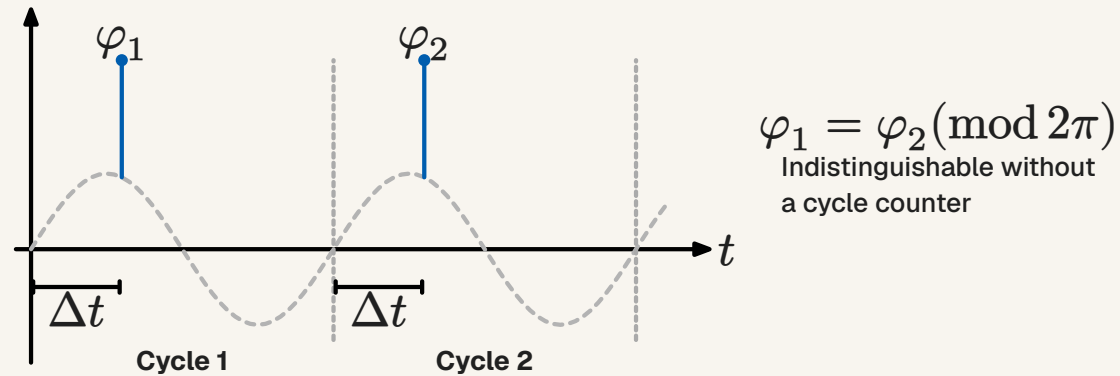


Rate encoded



TTFS encoded

TTFS PHASE AMBIGUITY



Spikes occurring at the same relative phase (φ_1 and φ_2) across different oscillation cycles are mathematically indistinguishable ($\varphi_1 = \varphi_2 \pmod{2\pi}$). Without a mechanism to track the global cycle count, downstream neurons cannot determine whether a spike represents a delayed response to a previous stimulus or an early response to a new one.

BIOLOGICAL NEURAL NETWORKS

Lateral Inhibition and Winner-Takes-All (WTA):

- An active excitatory neuron stimulates nearby inhibitory interneurons, which in turn suppress competing excitatory neighbours
- This engenders a **WTA dynamic** — the most active neuron silences its competitors, providing a physical mechanism for categorical decisions without a central processor

Excitation-Inhibition balance is critical:

- Excess excitation → runaway feedback loops (analogous to seizures)
- Excess inhibition → signal extinction (quiescence)

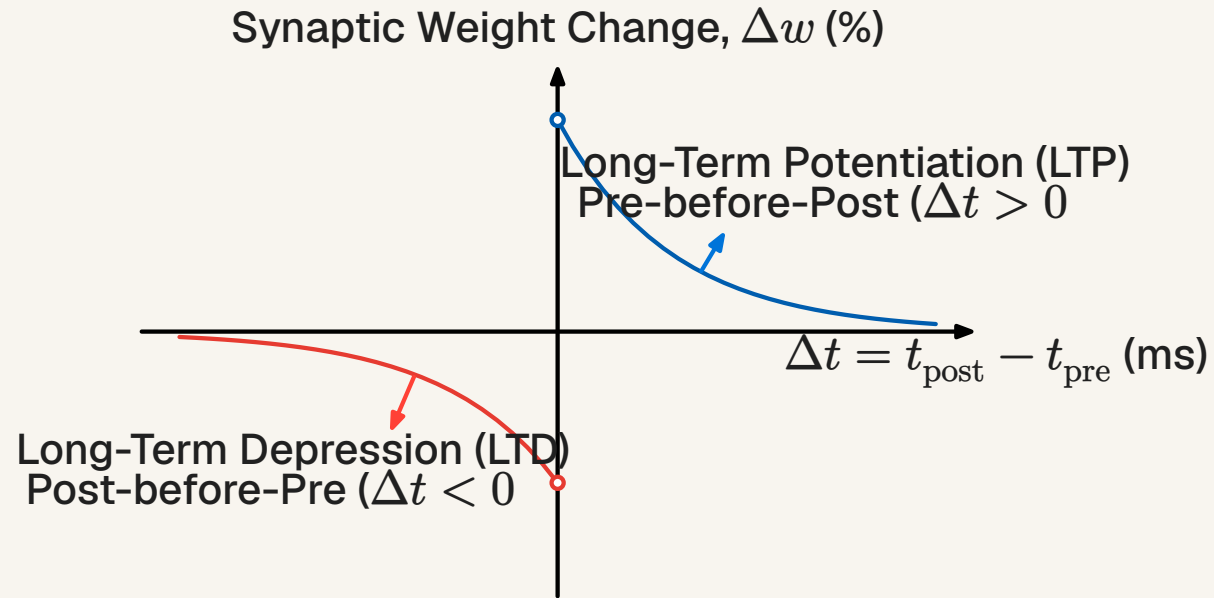
The brain operates at the critical point between these two extremes. Achieving this balance is essential in engineered SNNs.

BIOLOGICAL LEARNING

Spike-Timing-Dependent Plasticity (STDP):

A causal, millisecond-scale refinement of Hebbian learning. Weight update depends on the timing difference $\Delta t = t_{\text{post}} - t_{\text{pre}}$:

- **Long-Term Potentiation (LTP):** Pre fires **before** post ($\Delta t > 0$) — input contributed to firing → synapse **strengthened**
 - **Long-Term Depression (LTD):** Pre fires **after** post ($\Delta t < 0$) — input was irrelevant → synapse **weakened**
-
- Fully local: no global error signal required — only pre- and post-synaptic activity at the cleft
 - **Homeostatic plasticity** prevents runaway growth: if a neuron's average rate exceeds a target, all incoming weights are globally scaled down



BACK TO WHY DEEP LEARNING IS INEFFICIENT

Deep learning shares many core ideas with neuroscience but has diverged into its own discipline over time.

SIMILARITIES

- Deep learning has a version of neurons (the perceptron); it integrates and fires when reaching a threshold.
- It has neural networks—neurons connected together in hierarchical topologies to extract features.

DIFFERENCES

- **Signals:** ANNs use continuous, synchronous floating-point values (activations). The brain uses discrete, asynchronous events (spikes).
- **Optimization:** ANNs rely on backpropagation (global error gradients and exact weight transport). The brain relies on local plasticity (STDP).
- **Hardware Mapping:** ANNs execute dense matrix multiplications regardless of data sparsity. Biological networks communicate sparsely, saving massive amounts of energy.

BOTTLENECKS OF DEEP LEARNING

Standard deep learning imposes four compounding hardware bottlenecks:

- **Von Neumann Bottleneck:** Weights shuttle between memory and compute each step. Moving 32 bits from DRAM costs 640 pJ; computing with them costs 0.1 pJ — **Energy is wasted on transport, not computation**
- **Dense Processing of Sparse Data:** GPUs execute $0 \times w = 0$ multiplications for zero activations — structurally blind to sparsity, even when ReLU produces mostly-zero activation maps
- **Clock Synchrony Tax:** The global clock distribution network alone consumes 30–40% of chip power, even when circuits are idle
- **Backpropagation Locking:** All intermediate activations must remain in VRAM for the full backward pass; local synapses cannot adapt until the global error loop completes

NEUROMORPHIC PRINCIPLES

Three architectural pillars that directly address each bottleneck:

Co-location of Memory and Compute (The Synaptic Principle): Synaptic weights live with the neuron — zero data transport cost. Learning and inference are co-located.

Event-Driven Asynchrony (The Action Potential Principle): No global clock. Energy scales with task activity, not network size. Silent neurons consume nothing.

Sparse Binary Communication (The Spike Principle): Information is encoded in spike timing, not magnitude. Bandwidth scales with information content, not data dimensionality.

METHOD

To answer the research questions:

Sparse Efficient Computing: Investigate whether biologically inspired, event-driven algorithms reduce the computational footprint of visual classification when simulated on standard hardware. **Neuron Model Evaluation:** Identify which temporal integration dynamics are compatible with Time-To-First-Spike (TTFS) rank-order decoding across systematic threshold regimes. **Inference Via Weight Transfer:** Quantify the accuracy penalty of zero-shot Artificial-Neural-Network (ANN) to Spiking-Neural-Network (SNN) weight transfer under TTFS encoding, isolating the cost of transitioning to event-driven integration. **Native Unsupervised Learning:** Determine if local Spike-Time-Dependent-Plasticity (STDP) can autonomously extract meaningful geometric features from visual input without global error signals.

I propose the following experimental setup...

EXPERIMENT SETUP

The experiments are divided into three phases:

- **Phase I:** Neuron model evaluation — finding a suitable and efficient neuron model to use for Phase II and Phase III.
- **Phase II:** Weight transfer — using a pre-trained network via established methods and transferring weights to an SNN to investigate computational savings.
- **Phase III:** Training an SNN from scratch with zero dependence on traditional frameworks for potentially maximum efficiency.

DATASET AND NETWORK ARCHITECTURE

MNIST (28×28 normalized grayscale images), chosen for its high degree of spatial sparsity.

- **Topology:** Fully Connected Network (FCN) $784 \rightarrow 128 \rightarrow 10$
 - An FCN allows mathematically transparent, one-to-one parameter mapping without the structural overhead of convolutional unrolling
 - **No bias terms** — to preserve pure event-driven sparsity and simplify ANN→SNN transfer
- **SNN Simulator:** Custom PyTorch discrete-time engine operating over a $T_{\max} = 64$ tick saccade window
- **WTA:** Winner-Takes-All (WTA) — first output neuron to cross threshold claims the class label WTA is also used in the hidden layer during learning with STDP

PHASE I: NEURON MODEL EVALUATION

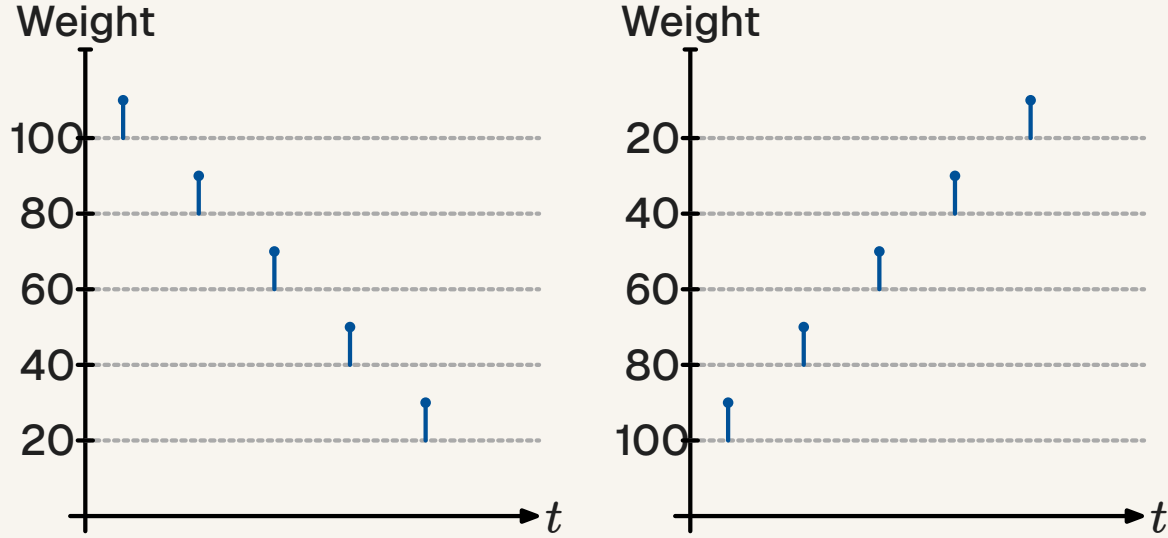
A **unit test** for isolated neuron temporal dynamics. Two synthetic spike trains are constructed:

- **Concordant:** Strongest synaptic weights arrive **first** — aligned with TTFS rank-order priority
- **Discordant:** Strongest weights arrive **last** — inverted temporal order

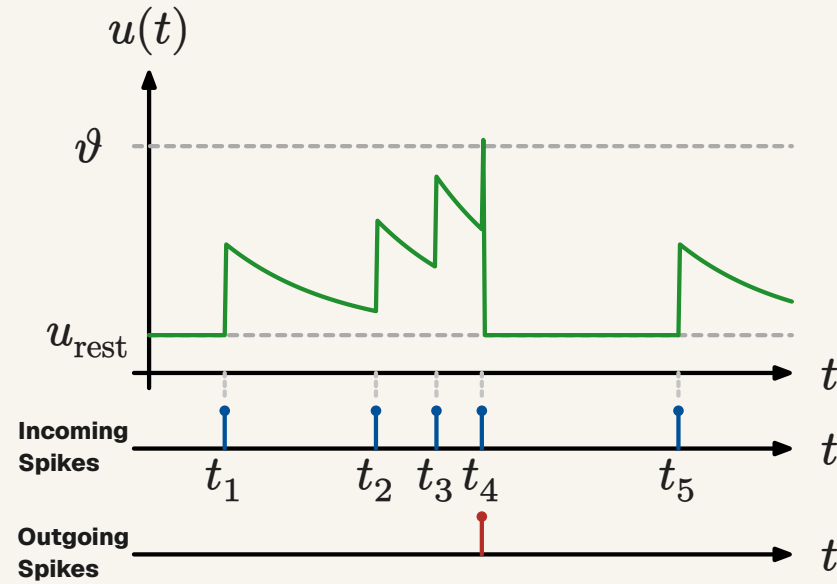
Each model is then evaluated across three threshold regimes:

- **Saturation (low ϑ):** Does the model fire early when evidence is abundant?
- **Critical (balanced ϑ):** Can the model discriminate **temporal order**, not just total magnitude?
- **Deficit (high ϑ):** Does the model fail gracefully when total weight is insufficient to cross the threshold?

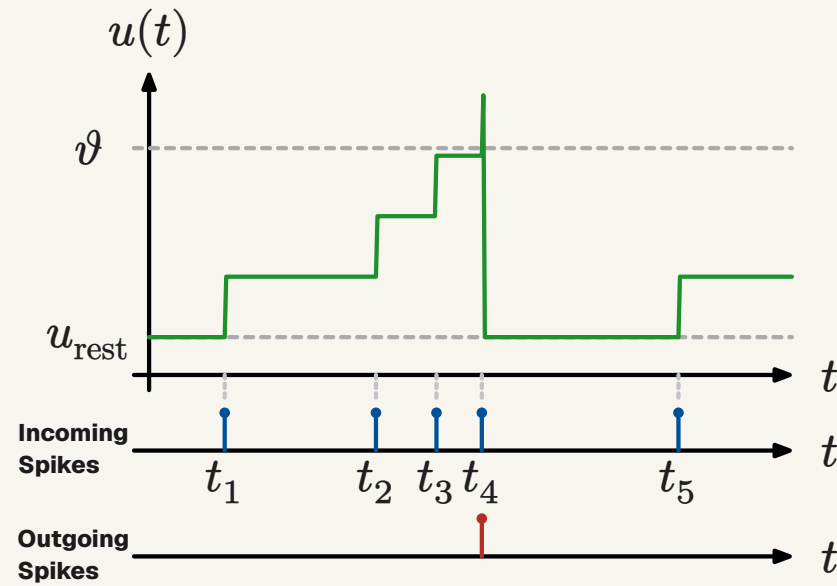
No MNIST data is used — this phase isolates the neuron's mathematical response to controlled spike patterns.



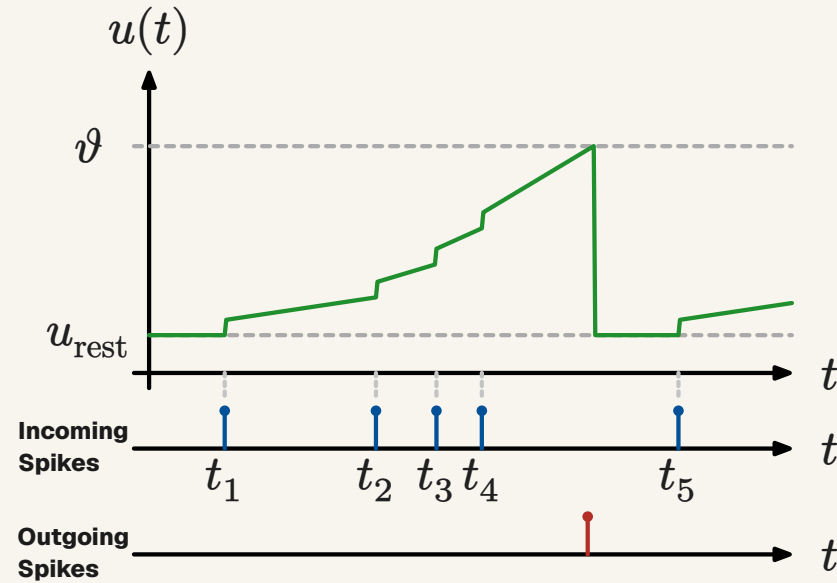
LIF MODEL



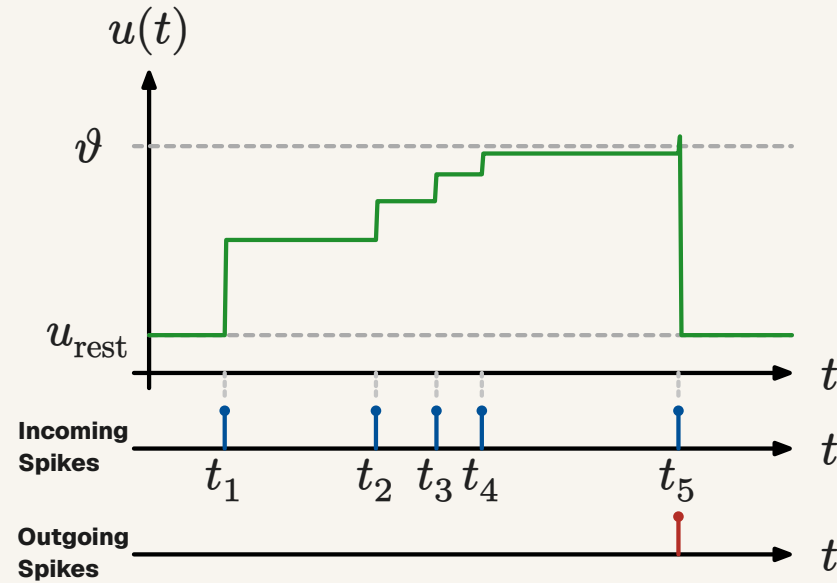
IF MODEL



LINEAR ACUMULATING RAMP MODEL



STATE DISCOUNT MODEL



PHASE II: ZERO-SHOT WEIGHT TRANSFER

1. Train a standard ANN baseline ($784 \rightarrow 128 \rightarrow 10$, Adam optimizer, 1000 epochs, no bias terms)
2. Copy FP32 weights directly to a structurally identical SNN — scaled by $\times 64$ to expand dynamic range for momentum integration
3. Run SNN inference using Model C over $T_{\max} = 64$ ticks with WTA output

Goal: Measure the **temporal penalty** — accuracy lost purely by switching from static continuous dot-product activations to discrete TTFS momentum integration, with **no retraining**.

Efficiency is tracked via SyOPs vs MACs and mean time-to-decision latency across the 10,000-image test set.

PHASE III: LEARNING

The SNN is trained **from scratch** — randomized weights, no labels, no backpropagation. Learning loop per image (one saccade):

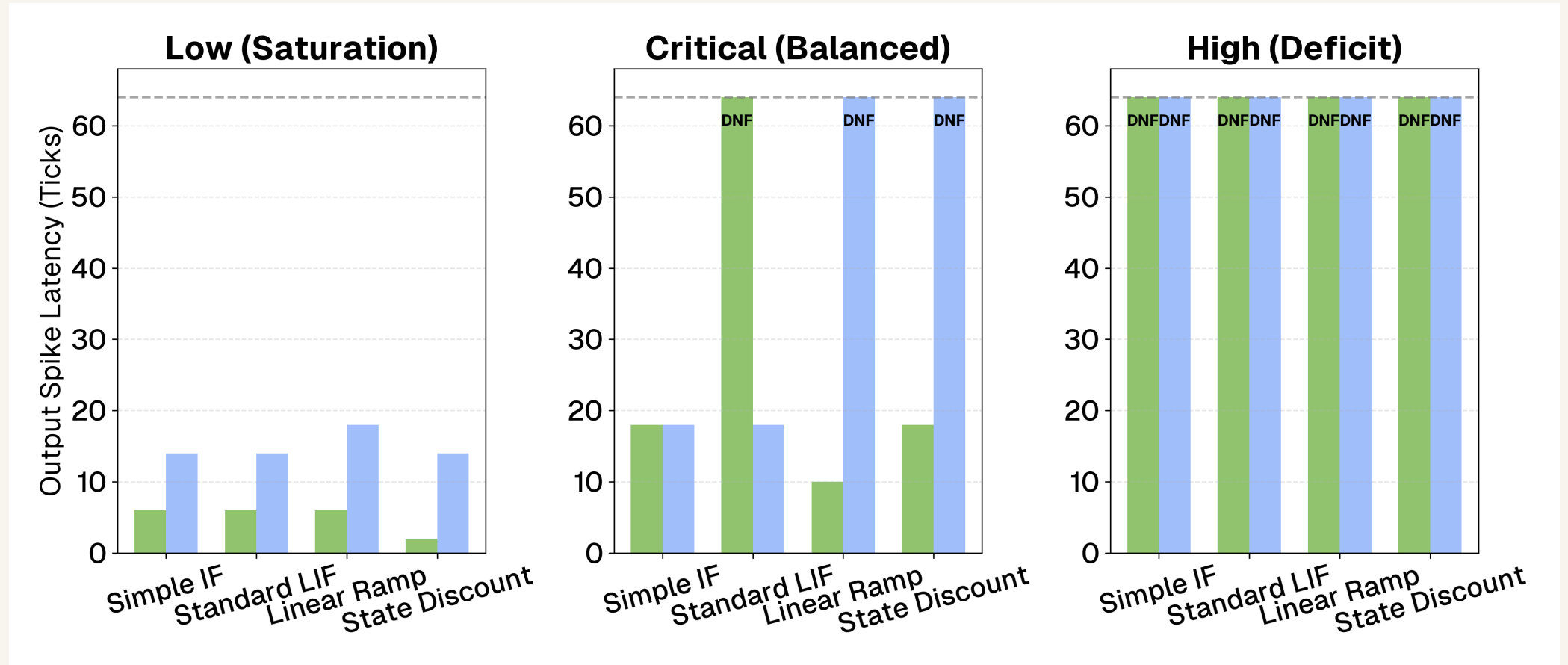
1. Encode image as TTFS spike train ($t_i = (1 - p_i) \cdot 64$, background suppressed)
2. Forward pass — all 128 hidden neurons integrate independently via Model C
3. **Post-hoc Hard WTA:** Only the single earliest-firing neuron is permitted to update
4. **Vectorized STDP:** Pre-spikes before the winner's time $\rightarrow$ LTP ($+A_+$); after or silent $\rightarrow$ LTD ($-A_-$)
5. **Homeostatic adaptation:** Winner receives threshold penalty ($+600$); all thresholds decay ($\times 0.90$)

Post-hoc labeling after training: pass labeled data through the frozen network; assign each output neuron the digit class it most frequently wins for. Then measure accuracy on the test set.

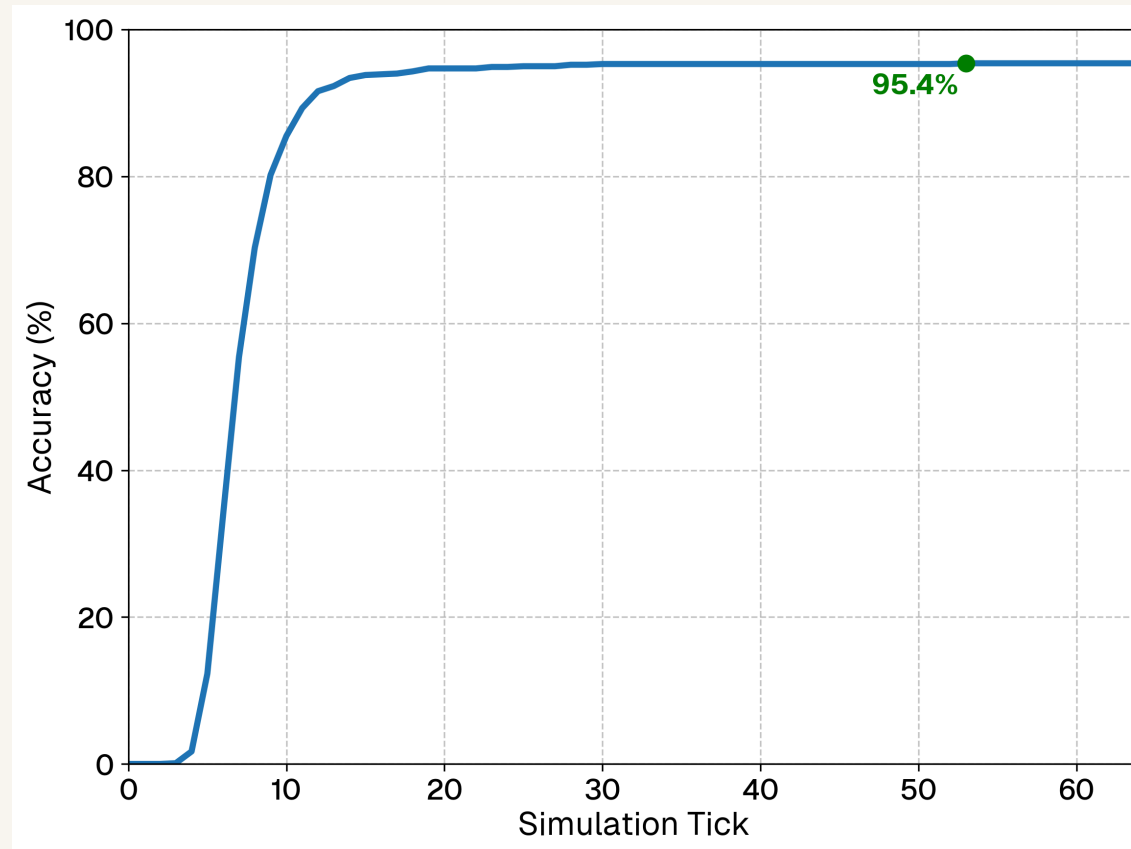
METRICS

- **Metrics:** Top-1 Accuracy, Temporal Latency (Time-to-Decision), and Synaptic Operations (SyOPs) as a hardware proxy

PHASE I: NEURON MODEL EVALUATION – RESULTS



PHASE II: ZERO-SHOT WEIGHT TRANSFER – RESULTS



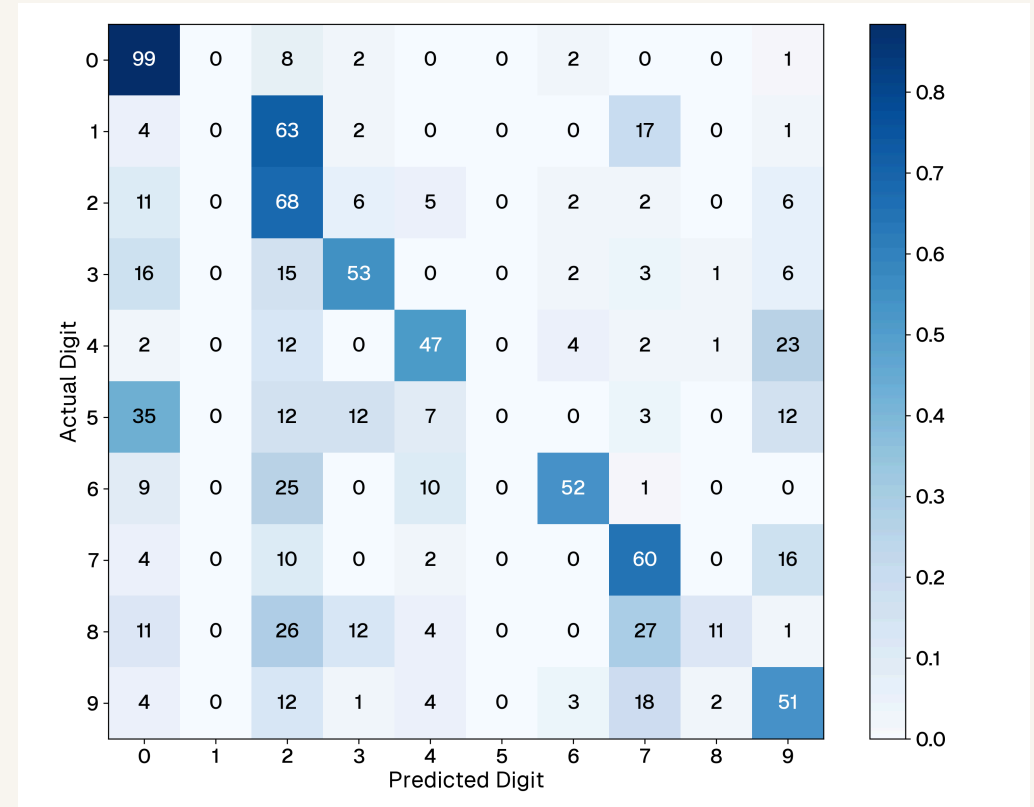
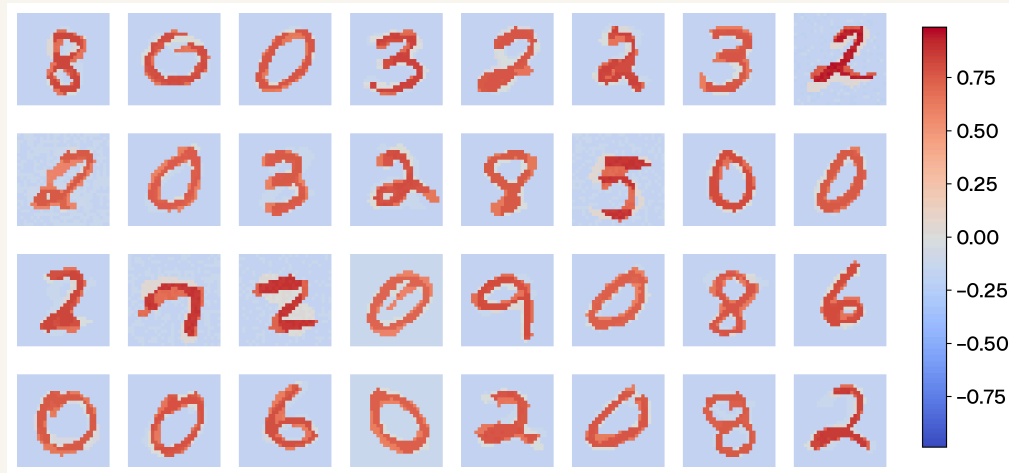
ANN baseline (1000 epochs, Adam, bias=false): **98.40%**

Zero-shot SNN (Model C, FP32 weights, ×64 scale): **94.50%** — temporal penalty of only **3.9 pp**

The S-curve profile confirms the **early-exit hypothesis**: >80% of correct decisions are locked in between ticks 5–15. Mean time-to-decision: **8.4 ticks** out of 64 → **85.2% reduction** in operations (15,021 SyOPs vs 101,632 MACs).

PHASE III: LOCAL SNN LEARNING – RESULTS

Unsupervised STDP + Hard WTA + homeostasis: **44.3% top-1 accuracy**



- Mean latency: **8.3 ticks** — comparable to supervised transfer
- SyOPs: **12,954** → **87.2% reduction** (marginally sparser than Phase II)

Hidden neurons self-organized into holistic digit templates without any labels. Visually simple digits ('0', '1') cluster reliably; morphologically ambiguous pairs ('5'/'0', '4'/'9') cause the most confusion — a structural limit of the single-layer WTA topology, not a failure of STDP.

DISCUSSION: KEY FINDINGS & LIMITATIONS

Neuron Model Evaluation:

- Classical LIF is fundamentally misaligned with TTFS — biological fidelity must be selectively adapted for engineering objectives
- Model C achieves temporal discriminability through purely linear mechanics, avoiding expensive exponential kernels

Weight Transfer Viability:

- 94.5% zero-shot accuracy proves ANN spatial hierarchies are recoverable from spike timing alone — hybrid pipelines are viable for edge deployment
- **Sparsity Paradox:** GPUs still execute boolean-masked zero multiplications; physical efficiency gains require native neuromorphic ASICs or FPGAs

Unsupervised Learning:

- STDP clusters geometric features without labels — the 44.3% result is not a failure of the learning rule but of the single-layer fully-connected WTA topology
- Holistic template matching breaks down on morphologically overlapping digit classes

FUTURE WORK

- **Native temporal datasets:** N-MNIST, DVS Gesture sensors — output spike trains natively, eliminating the TTFS encoding bottleneck entirely
- **Spiking CNN architectures:** Local receptive fields + local WTA → neurons learn edge and curve primitives rather than rigid full-digit templates, enabling translation invariance
- **Structural plasticity:** Prune consistently depressed synapses → block-sparse tensor formats → real memory bandwidth reductions on FPGAs and ASICs
- **Hardware deployment:** Port verified TTFS algorithms to Intel Loihi or IBM TrueNorth to measure true joule-per-inference energy consumption

Analog crossbar arrays (memristors, spintronics) represent the ultimate physical substrate — co-locating synaptic weights with computation in a single analog device, realizing the full in-memory computing vision.

CONCLUSION

The brain runs on **20 W**. Modern AI clusters run on **megawatts**. Closing this gap is a paradigmatic challenge — not an engineering detail.

This thesis demonstrated:

- **TTFS encoding** reduces theoretical compute by 85–87% vs. dense ANN inference via temporal early-exit sparsity
- **Standard LIF fails** at TTFS decoding — Model C (current-accumulating linear ramp) is a necessary correction
- **Zero-shot weight transfer** recovers 94.5% accuracy with only a 3.9 pp temporal penalty — no retraining required
- **Unsupervised STDP** successfully extracts geometric features (44.3%) without labels or backpropagation

The “Sparsity Paradox” — GPUs executing masked zero multiplications — confirms that **realizing these efficiency gains requires native neuromorphic substrates**.

The majority of the teoretical savings come from the TTFS encoding and the challange is building systems around this

The algorithmic foundations are proven. The remaining challenge is hardware. As Intel Loihi, IBM TrueNorth, and analog crossbar arrays mature, the 20-watt brain becomes the blueprint — not just the metaphor.

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